

CO1 - Assessment tool 1:-

1) Full-duplex voice channels:-

Given:-

$$\text{Total Spectrum} = 45\text{MHz} = 45,000\text{KHz.}$$

$$\text{Simplex channel} = 25\text{KHz.}$$

$$\text{Full-duplex channel} = 2 \text{ Simplex channels.}$$

$$\text{Full-duplex channels} = \frac{\text{Total Spectrum}}{2 \times \text{Bandwidth per Simplex channel.}}$$

$$= \frac{45000}{2 \times 25} = \frac{45000}{50} = 900$$

2) channels per cell:-

$$\text{channels per cell} = \frac{\text{Total voice channels}}{\text{Number of cells in cluster}}$$

$$\text{Total channels} = 560.$$

$$\text{cluster size} = 7.$$

$$= \frac{560}{7} = 80.$$

3.) Total simultaneous voice channels.

Given: carriers = 100

Time Slots = 8

Total voice channels = Number of carriers $\times$ Time Slot per carrier.

$$= 100 \times 8$$

$$= 800$$

4.) power gap in dB:-

Given:-

Base Station Power = 80W.

Handset Power = 4W.

$$P_{dB} = 10 \log_{10} \left(\frac{P_1}{P_2} \right)$$

$$\Rightarrow 10 \log_{10} \left(\frac{80}{4} \right) = 10 \log_{10} (20)$$

$$= 10 \log_{10} (1.301) = 13.01 \text{ dB.}$$

5.) Carrier - to - interference ratio:-

given:-

carrier signal = -60 dBm.

Interference = +92 dBm.

$$C/I \text{ (dB)} = C \text{ (dBm)} - I \text{ (dBm)}.$$

$$= (-60) - (-92)$$

$$= -60 + 92$$

$$= 32 \text{ dB.}$$

6) channels per cell

given:

$$\text{Total channels} = \frac{\text{Total bandwidth}}{\text{channel bandwidth}}$$

$$= \frac{42 \text{ MHz}}{300 \text{ KHz}}$$

convert 42 MHz to KHz:-

$$42 \text{ MHz} = 42000 \text{ KHz}$$

$$= \frac{42000}{300} = 140 \text{ channels}$$

Formula 2:-

$$\text{channels per cell} = \frac{\text{Total channels}}{\text{cluster size}}$$

$$= \frac{140}{7} = 20$$

7) Received Power after doubling distance:-

$$P_2 = P_1 \left(\frac{d_1}{d_2} \right)^n$$

$$P_1 = 64 \text{ W}$$

$$d_2 = 2d_1$$

Path-loss exponent, $n = 4$

$$P_2 = 64 \left(\frac{d_1}{2d_1} \right)^4$$

$$= 64 \left(\frac{1}{2} \right)^4 = 64 \times \frac{1}{16}$$

$$= 4 \text{ W}$$

8) Power required when radius doubles :-

$$\frac{P_2}{P_1} = \left(\frac{d_2}{d_1} \right)^n$$

$$n = 2$$

$$d_2 = 2d_1$$

$$\frac{P_2}{P_1} = \left(\frac{2d_1}{d_1} \right)^2 = 2^2 = 4$$

9) Total channel count in cluster :-

$$\text{cluster size} = 4$$

$$\text{channels per cell} = 15$$

$$\text{Total channels} = \text{cluster size} \times \text{channels per cell}$$

$$= 4 \times 15 = 60$$

10) Doppler shift :-

$$f_d = \frac{v f_c}{c}$$

$$v = 100 \text{ km/h}$$

$$f_c = 900 \text{ MHz}$$

$$c = 3 \times 10^8 \text{ m/s}$$

$$100 \text{ km/h} = \frac{100 \times 1000}{3600} = 27.78 \text{ m/s}$$

$$f_d = \frac{27.78 \times 900 \times 10^6}{3 \times 10^8} = 83.3 \text{ Hz.}$$

11) TDMA Slot duration:-

$$\text{Slot duration} = \frac{\text{Frame duration}}{\text{No. of time slots.}}$$

$$\text{Frame duration} = 5 \text{ ms} \left(\frac{1}{1000} \right) = \frac{5}{1000}$$

$$\text{Number of slots} = 8.$$

$$= \frac{5}{8} = 0.625 \text{ ms.}$$

12) Number of carrier frequencies:-

$$\text{Number of carriers} = \frac{\text{Total bandwidth}}{\text{carrier spacing.}}$$

$$\text{Bandwidth} = 24 \text{ MHz} = 24,000 \text{ KHz.}$$

$$\text{carrier spacing} = 200 \text{ KHz.}$$

$$= \frac{2400}{200} = 120.$$

13) Received Power in dBm:-

convert watts to dBm:-

$$P(\text{dBm}) = 10 \log_{10} (P(\text{W}) \times 1000).$$

$$P = 8 \text{ W.}$$

$$P(\text{dBm}) = 10 \log_{10} (8 \times 1000).$$

$$= 10 \log_{10} (8000) = 39.03 \text{ dBm.}$$

Received Power :-

$$P_{r1} = P_t - \text{Path loss}$$

$$= 39.03 - 33$$

$$= 6.03 \text{ dBm}$$

4.) Free-Space Path loss :-

$$\text{FSPL (dB)} = 32.44 + 20 \log_{10}(d \text{ km}) + 20 \log_{10}(f \text{ MHz})$$

$$\text{Distance} = 2.5 \text{ km}$$

$$\text{Frequency} = 1900 \text{ MHz}$$

$$\text{FSPL} = 32.44 + 20 \log_{10}(2.5) + 20 \log_{10}(1900)$$

$$= 32.44 + 7.96 + 65.58$$

$$= 105.98 \text{ dB}$$

15.) cluster call capacity :-

$$\text{cluster capacity} = \text{Number of cells} \times \text{calls per cell}$$

$$\text{cells} = 6$$

$$\text{calls per cell} = 250$$

$$= 6 \times 250$$

$$= 1500$$

16.) Active calls :-

$$N_a = \lambda \times T$$

$$\lambda = 4000 \text{ calls/sec}$$

$$T = 3 \text{ minutes}$$

convert min to Secs :-

$$3 \times 60 = 180 \text{ Seconds}$$

$$N = 4000 \times 180$$

$$= 720000$$

17) Total bandwidth:-

$$\text{channels} = 150$$

$$\text{Each channel} = 25 \text{ KHz}$$

$$\text{Total bandwidth} = \text{Number of channels} \times \text{Bandwidth per channel}$$

$$= 150 \times 25$$

$$= 3750 \text{ KHz}$$

convert to MHz:-

$$\frac{3750}{1000} = 3.75 \text{ MHz}$$

18) Total guard-band bandwidth:-

Formula 1:-

$$\text{Number of guard bands} = \text{Number of frequency bands} - 1$$

$$= 30 - 1 = 29$$

Formula 2:-

$$\text{Wasted bandwidth} = \text{Number of guard bands} \times \text{Guard band size}$$

$$= 29 \times 40 \text{ KHz}$$

$$= 1160 \text{ KHz}$$

$$= 1.16 \text{ MHz}$$

19.) Number of handoffs per hour:-

cell diameter:-

$$D = 2R.$$

$$D = 2 \times 1.5 = 3 \text{ km}.$$

Handoffs:-

$$\text{Handoffs} = \frac{\text{Distance traveled}}{\text{cell diameter}}.$$

$$= \frac{90 \text{ km}}{3 \text{ km}} = 30.$$

20.) Total user-handling capacity:-

$$\text{Total users} = \text{Regional centers} \times \text{Base}$$

$$\text{Stations per center} \times \text{user per base station}.$$

$$= 8 \times 40 \times 250.$$

$$= 320 \times 250 = 80000.$$

21.) Number of cells required:-

$$\text{Number of cells} = \frac{\text{Total area}}{\text{Area of one cell}}.$$

$$\text{Total area} = 400 \text{ km}^2.$$

$$\text{Radius} = 400 \text{ m} = 0.4 \text{ km}.$$

$$\text{Approximate cell area} = R^2.$$

$$\text{cell area} = (0.4)^2.$$

$$= 0.16 \text{ km}^2.$$

$$\text{Number of cells} = \frac{400}{0.16} = 2500$$

22) Number of FDMA channels:-

$$\text{Number of channels} = \frac{\text{Total Spectrum}}{\text{channel bandwidth}}$$

$$\text{Total Spectrum} = 6\text{MHz} = 6000\text{KHz}$$

$$\text{channel bandwidth} = 30\text{KHz}$$

$$= \frac{6000}{30} = 200$$

23) offered traffic in erlangs:-

$$A = N \times \lambda \times H$$

$$N = 1200$$

$$\text{calls / hr} = 4$$

$$\text{call duration} = 2.5 \text{ minutes}$$

$$H = \frac{2.5}{60}$$

$$= 0.04167 \text{ hrs}$$

$$A = 1200 \times 4 \times 0.04167$$

$$= 200 \text{ Erlangs}$$

24) Maximum number of users:-

$$\text{Number of users} = \frac{\text{Total bandwidth}}{\text{Bandwidth per user}}$$

$$\text{Total bandwidth}$$

$$= 40\text{MHz} = 40,000\text{KHz}$$

$$\text{user bandwidth} = 400\text{KHz}$$

$$= \frac{40000}{400} = 100$$

25) Total network user capacity:-

Total capacity = Number of base Station $\times$ user per base Station.

$$= 1200 \times 600 = 720000$$

26) Paging rounds required:-

$$\text{Paging rounds} = \frac{\text{Total cells}}{\text{cells searched per round}}$$

$$= \frac{200}{4} = 50$$

27) Fraction of offered traffic blocked:-

$$A = \lambda \times T$$

$$\text{call rate} = 40000 \text{ calls/min}$$

$$\text{Average duration} = 3 \text{ mins}$$

$$A = 40000 \times 3$$

$$= 120000 \text{ Erlangs}$$

$$C = 100000$$

$$\text{Blocking fraction} = \frac{A - C}{A}$$

$$= \frac{120000 - 100000}{120000} = \frac{20000}{120000}$$

$$= 0.1667$$

convert to %:-

$$0.1667 \times 100 = 16.67\%$$

28) Subscribers roaming:-

$$600000 \times \frac{15}{100}$$

$$= 600000 \times 0.15$$

$$= 90000$$

29) Percentage of capacity utilized:-

Utilization Percentage =

$$\frac{\text{current load}}{\text{maximum capacity}} \times 100$$

$$= \frac{9000}{12000} \times 100$$

$$= 0.75 \times 100 = 75\%$$

30) channels per cell:-

$$\text{Total channels} = \frac{\text{Total Spectrum}}{\text{channel bandwidth}}$$

$$= \frac{48 \text{ MHz}}{200 \text{ kHz}}$$

convert:-

$$48 \text{ MHz} = 48000 \text{ kHz}$$

$$= \frac{48000}{200} = 240 \text{ channels}$$

channels per cell :-

$$\text{channels per cell} = \frac{\text{Total channels}}{\text{Reuse factor}}$$

$$= \frac{240}{4} = 60$$

31) GSM channels :-

$$\text{Number of channels} = \frac{\text{Total bandwidth}}{\text{channel spacing}}$$

$$\text{Bandwidth} = 30 \text{ MHz} = 30,000 \text{ KHz}$$

$$\text{channel spacing} = 200 \text{ KHz}$$

$$= \frac{30000}{200} = 150$$

32) Successful handoffs :-

$$\text{Successful handoffs} = \text{Total handoffs} \times \text{Success rate}$$

$$= 2500 \times \frac{99}{100}$$

$$= 2500 \times 0.99$$

$$= 2475$$

33) Received Power at base station :-

First convert 12W to dBm

$$P(\text{dBm}) = 10 \log_{10} (P(\text{W}) \times 1000)$$

$$= 10 \log_{10} (12 \times 1000)$$

$$= 10 \log_{10} (12000)$$

$$= 40.79 \text{ dBm}$$

Formula 2 :-

$$P_{rx} = P_t - \text{Path loss}$$

$$= 40.79 - 45$$

$$= -4.21 \text{ dBm}$$

34) cluster user capacity:-

$$\text{cluster capacity} = \text{Number of cells} \times \text{users per cell}$$

$$= 7 \times 900$$

$$= 6300$$

35) TDMA Slot duration:-

$$\text{Slot duration} = \frac{\text{Frame duration}}{\text{Number of Slots}}$$

$$= \frac{4.8 \text{ ms}}{8} = 0.6 \text{ ms}$$

36) Doppler shift:-

$$f_d = \frac{v f_c}{c}$$

convert velocity :-

$$150 \text{ Km/h} = \frac{150 \times 1000}{3600}$$

$$= 41.67 \text{ m/s}$$

convert frequency:-

$$1.8 \text{ GHz} = 1.8 \times 10^9 \text{ Hz}$$

$$f_d = \frac{41.67 \times 1.8 \times 10^9}{3 \times 10^8}$$

$$= 250 \text{ Hz.}$$

37) Share of offered load blocked:-

$$A \times A = \lambda \times T.$$

$$A = 70000 \times 4.$$

$$= 280000 \text{ Erlangs.}$$

capacity :-

$$C = 250000$$

$$\text{Blocking fraction} = \frac{A - C}{A}.$$

$$= \frac{280000 - 250000}{280000}.$$

$$= \frac{30000}{280000}$$

$$= 0.10714.$$

Percentage :-

$$0.10714 \times 100.$$

$$= 10.71\%.$$

38) cluster capacity :-

$$\text{capacity} = \text{cells} \times \text{users per cell.}$$

$$= 9 \times 600 = 5400 \text{ users.}$$

39.) Dropped calls per Second:-

$$\text{Dropped calls} = \text{Total calls} \times \text{Drop rate.}$$

$$= 60000 \times 0.004 = 240 \text{ calls/sec.}$$

40.) Path Loss:-

$$PL = P_t - P_r.$$

$$= 43 - (-81) = 130 \text{ dB.}$$

41.) Total geographic coverage:-

$$\text{Total area} = \frac{\text{Area per Base Station} \times \text{Number of Stations}}{A}$$

$$= 2.5 \times 150 = 375 \text{ Km}^2.$$

42.) Channels per cell:-

$$\text{Total channels} = \frac{\text{Total Spectrum.}}{\text{channel Bandwidth.}}$$

$$= \frac{48 \times 1000}{200} = 240.$$

$$\text{channels per cell} = \frac{240}{6} = 40.$$

43.) Paging Cycles:-

$$\text{Paging cycles} = \frac{\text{Total users.}}{\text{users Per cycle.}}$$

$$= \frac{150000}{6000} = 25$$

44) Number of handoffs =

$$\text{Handoffs} = \frac{\text{call duration}}{\text{Handoff interval}}$$

$$= \frac{9}{3} = 3$$

45) Signal to noise ratio:-

$$\text{SNR (dB)} = P_{\text{signal}} - P_{\text{noise}}$$

$$= (-85) - (-115)$$

$$= 30 \text{ dB}$$

46) Minimum bandwidth:-

$$= 250 \times 150 \text{ Kbps}$$

$$= 37500 \text{ Kbps}$$

$$= 37.5 \text{ Mbps}$$

47) Traffic in erlangs:-

$$A = N \times C \times T$$

convert call duration:-

$$T = \frac{2.5}{60} = 0.04167 \text{ hour}$$

$$A = 1500 \times 4 \times 0.04167$$

$$= 250 \text{ Erlangs}$$

48.) control channel bandwidth:-

$$\begin{aligned}\text{Total Bandwidth} &= \text{No. of channels} \times \text{Bandwidth per channel} \\ &= 250 \times 12 \text{ KHz} \\ &= 3000 \text{ KHz} \\ &= 3 \text{ MHz}\end{aligned}$$

49.) Total throughput:-

$$\begin{aligned}\text{Total throughput} &= \text{no. of connections} \times \text{Throughput per connection} \\ &= 1500 \times 600 \text{ Kbps}\end{aligned}$$

$$\begin{aligned}&= 900000 \text{ Kbps} \\ &= 900 \text{ Mbps}\end{aligned}$$

50.) Number of towers:-

Area of one tower:-

$$\begin{aligned}A &= \pi R^2 \\ &= \pi (4)^2 \\ &= 16\pi \approx 50.27 \text{ Km}^2\end{aligned}$$

Number of towers:-

$$\begin{aligned}N &= \frac{\text{Total Service area}}{\text{Area covered by one tower}} \\ &= \frac{350}{50.27} = 6.96\end{aligned}$$

Round up to the next whole tower = 7.